



CMSC 449 – Fall 2025
Syllabus v2

Section 1: Course Information

Course Number	CMSC 449	
Course Name	Malware Analysis	
Locations	CMSC 449-01 CMSC 449-100	ITE 233 Online
Term	Fall 2025	
Instructor	RJ Joyce	joyce8@umbc.edu
TA	Sree Sai Bhargav Sarvepalli	yj49411@umbc.edu
Course Website	BlackBoard	
Office Hours	RJ: Sree Sai:	Mo/We 1:30pm - 3:00pm TBA
Textbook	The textbook Practical Malware Analysis (Sikorski and Hoenig) is required for this course.	

All lectures will be recorded, and you may attend class in-person or via WebEx. The entire course can be completed remotely, if preferred.

WebEx Link:

- <https://umbc.webex.com/umbc/j.php?MTID=mae3d320454b6c34b879c24a1f3b1302a>

Lecture Recordings:

- <https://drive.google.com/drive/folders/1f8gO-TdFjdutHiNu5FsxskYVC7Es2END?usp=sharing>

Office Hours Information:

- **RJ:** Mo/We 1:30pm - 3:00pm in ITE 329-A. Also available to meet via WebEx or at other times by request.
- **Bhargav:** Fridays 1:00pm – 3:00pm on Discord

Course Communications:

This course will use Blackboard for hosting homework assignments, course materials, and grades. It is your responsibility to check it often. Important announcements will be posted on Blackboard, sent via email, and posted to the #announcements channel on the course Discord server. Minor announcements (e.g. homework corrections and clarifications) will only be posted to the Discord server.

From 9am-5pm on weekdays, I am most reachable via email. I will also respond to emails and Discord messages during the evenings and weekends when available. I try to respond quickly, but please keep in mind that I have other work responsibilities. I often get a lot of questions about homeworks the day they are due, so I may be slower to respond on those days too. It's always best to start assignments early and ask questions before the last minute!

Course Discord Server:

You are encouraged to join the Discord server for this course. Students may use this for course discussion, but are prohibited from sharing code or answers to assignments.

- Server invite link: <https://discord.gg/d3Nmrxfj6e>

Analysis VM:

We will use a Windows 7 VM that comes with a multitude of pre-installed malware analysis tools. This VM will be required for completing nearly every assignment in the class, including the take-home midterm and final exams.

- VirtualBox download:
 - <https://www.virtualbox.org/wiki/Downloads>
- Windows 7 VM .ova file:
 - <https://drive.google.com/file/d/1wUy9elzAedZNXcfM0vVo0HliaFdGboUT/view?usp=sharing>

You must be logged into your UMBC account in order to access the .ova file on Google Drive. You will need appx. 25 GB of disk storage free to import the VM, not counting the size of the .ova file itself. (You may delete the .ova file once the VM has been imported).

Please try to download and import the VM early to ensure that it works properly. You will need a computer with virtualization support and that is compatible with Virtualbox. **Macs with Apple Silicon chips (M1, M2, M3, M4) are not compatible with the malware analysis VM.**

Section 2: Course Overview

We explore both static and dynamic malware analysis in this course. Although malware takes many forms, the focus will be on Windows executable files. We will cover a variety of tools used to investigate file properties, identify file behaviors, and dissect malicious code. A significant amount of the course will focus on the use of disassemblers, decompilers, and debuggers for reverse-engineering malware. This will involve reading x86 assembly and C code, and we will spend class time reviewing both languages. The class will also cover specialized topics such as process injection, handling packed or obfuscated malware, and writing signatures to detect malware.

Section 3: Tentative Course Outline:

This is a planned schedule of topics for each class period. Note that this schedule is subject to change as we progress through the semester.

Topic
Lec01 – Intro to Malware Analysis
Lec02 – Basic Static Analysis
Lec03 – Hashing in Malware Analysis
Lec04 – Packed Malware
Lec05 – VMs and Sandboxes
Lec06 – Host-Based Dynamic Analysis Tools
Lec07 – Network-Based Dynamic Analysis Tools
Lec08 – x86 Assembly Review
Lec09 – Identifying C Code Constructs in Assembly
Lec10 – IDA Pro
Lec11 – Ghidra
Lec12 – Malware and the Windows API
Lec13 – Convert Malware Launching
Lec14 – Immunity Debugger
Lec15 – Analyzing Shellcode

Lec16 – Manual Unpacking
Lec17 – YARA Rules
Lec18 – Data Encoding

There will also be participation exercises interspersed with these lectures. You will be given at least 1 week of notice of these via announcements on Blackboard, email, and Discord.

Section 4: Grading Criteria

Course grades will be a function of your performance in the following categories:

Homeworks	70%
Participation	15%
Final Exam	15%

Assignments: There are 8 planned homework assignments:

- HW 1: Basic Static Analysis
- HW 2: Basic Dynamic Analysis
- HW 3: x86 Assembly
- HW 4: Advanced Static Analysis with IDA Pro
- HW 5: Advanced Static Analysis with Ghidra
- HW 6: Advanced Dynamic Analysis
- HW 7: Unpacking Malware
- HW 8: YARA Rules

Participation: Polls will be used during class to encourage participation. They may be completed online or in-person. For each poll question, you will receive 1 point for answering correctly, ½ point for answering incorrectly, and 0 points for not answering. Other assignments may also count towards the participation grade, such as written responses to special topics lectures. At the end of the semester, the Participation grade will be curved by 20% to account for occasional absences.

Final Exam: The final exam will be a take-home assignment. You will be given multiple days to analyze 1-2 malicious files and answer questions about them using analysis tools/techniques taught in class. Questions will typically be similar to those asked on prior homework assignments, or will be related to topics demonstrated in class.

Final grades will be determined based on your weighted percentage:

A	$\geq 90\%$
B	80% - 89%
C	70% - 79%
D	60% - 69%
F	$< 60\%$

Section 5: Course Policies

Late policy: 10% of the total points will be deducted for each day that an assignment is late.

Attendance: You are expected to attend lectures (in-person or via WebEx) when possible. Although lecture attendance is not a direct component of your grade, students who attend class generally perform more highly than their non-attending peers. Viewing just the lecture slides is **NOT** a substitute for attending class, and you may be responsible for material covered during class that is not on the slides. Lecture recordings will be posted, but attending class in real-time is preferred.

Section 6: Academic Integrity

Academic dishonesty of any kind will be handled in accordance with University policy.

"By enrolling in this course, each student assumes the responsibilities of an active participant in UMBC's scholarly community, in which everyone's academic work and behavior are held to the highest standards of honesty. Cheating, fabrication, plagiarism, and helping others to commit these acts are all forms of academic dishonesty, and they are wrong. Academic misconduct could result in disciplinary action that may include, but is not limited to, suspension or dismissal. To read the full Student Academic Conduct Policy, consult the UMBC Student Handbook, the Faculty Handbook, or the UMBC Policies section of the UMBC Directory." [Statement adopted by UMBC's Undergraduate Council and Provost's Office.]

For this class, if you use ChatGPT (or similar chatbots or AI-based generation tools), you must describe exactly how you used it, including providing the prompt, original generation, and your edits. This applies to prose, code, or any form of content creation. Not disclosing is an academic integrity violation. If you do disclose, your answer may receive anywhere from 0 to full credit, depending on the extent of substantive edits, achievement of learning outcomes, and overall circumvention of those outcomes. Use of AI/automatic tools for grammatical assistance (such as spell-checkers or Grammarly) or small-scale predictive text (e.g., next word prediction, tab completion) is okay. Provided the use of these tools does not change the substance of your work, use of these tools may be, but is not required to be, disclosed.

The following are examples of academic integrity violations in this course:

- Using ChatGPT to complete your homework without following the instructions above
- Copying code or answers from another student

- Sharing code or answers with another student
- Using homework-help sites (such as Chegg and CourseHero)

Any academic integrity violations will result in an immediate zero on the assignment. A second academic integrity violation (or a violation on the midterm or final exam) will constitute an immediate failure in the course.

Section 7: Abuse of Resources

Abuse of the knowledge or experience you gain in this course may subject you to discipline under UMBC policy and/or criminal prosecution. Do not expect your status as a student to protect you if you break the law! Hacking into campus computers (other than systems approved for such a purpose) is a violation of UMBC policy, and may result in disciplinary action possibly including expulsion, in addition to possible criminal charges.

Section 8: Students with Accommodations

UMBC is committed to eliminating discriminatory obstacles that may disadvantage students based on disability. UMBC complies with federal legislation for individuals with disabilities (Section 504 of the Rehabilitation Act of 1973, the Americans with Disabilities Act of 1990, and the ADAA of 2009) that offers reasonable accommodations to qualified students with disabilities. Student Disability Services (SDS), formerly Student Support Services, is the UMBC department designated to:

- receive and maintain confidential files of disability-related documentation,
- certify eligibility for services,
- determine reasonable accommodations,
- develop with each student plans for the provision of such accommodations, and
- serve as a liaison between faculty members and students regarding disability-related issues.

If you have a documented disability and need to request accommodations, please refer to the SDS website at <http://sds.umbc.edu> or contact the office by phone at 410-455-2459, via email at disability@umbc.edu, or in person in Math/Psychology Room 212. If you require accommodations for this class, make an appointment to meet with your instructor to discuss your SDS-approved accommodations.

Section 9: Title IX

As an instructor, I am considered a Responsible Employee, per UMBC's Policy on Prohibited Sexual Misconduct, Interpersonal Violence, and Other Related Misconduct (located at <http://humanrelations.umbc.edu/sexual-misconduct/umbc-resource-page-for-sexual-misconduct-and-other-related-misconduct/>). While my goal is for you to be able to share information related to your life

experiences through discussion and written work, I want to be transparent that as a Responsible Employee I am required to report disclosures of sexual assault, domestic violence, relationship violence, stalking, and/or gender-based harassment to the University's Title IX Coordinator.

As an instructor, I also have a mandatory obligation to report disclosures of or suspected instances of child abuse or neglect (www.usmh.usmd.edu/regents/bylaws/SectionVI/VI150.pdf).

The purpose of these reporting requirements is for the University to inform you of options, supports and resources; you will not be forced to file a report with the police. Further, you are able to receive supports and resources, even if you choose to not want any action taken. Please note that in certain situations, based on the nature of the disclosure, the University may need to take action.

If you need to speak with someone in confidence about an incident, UMBC has the following Confidential Resources available to support you:

- The Counseling Center: 410-455-2472
- University Health Services: 410-455-2542
- (After-hours counseling and care available by calling campus police at 410-455-5555)

Other on-campus supports and resources:

- The Women's Center (available to students of all genders): 410-455-2714 (Mon-Thurs 9:30-6, Fri 9:30-4:00)
- Title IX Coordinator: 410-455-1606 (9-5)

Child Abuse and Neglect:

- Please note that Maryland law requires that I report all disclosures or suspicions of child abuse or neglect to the Department of Social Service and/or the police.